

MISSION 5

Part I. Problem 1 : What we learned from Zeeman's paper

Reference : E.C. Zeeman , *The umbilic bracelet and the double-cusp catastrophe* , Springer LNM 525 (1976) , pp. 328–366. (Free PDF at LMS Zeeman Archive.)

The skills and techniques we learned :

i. (Lemma 3 in the paper P-334) Every real cubic form in (x, y) can be written in the complex form :

$$f = \operatorname{Re}(\alpha z^3 + \beta z^2 \bar{z}) , \quad (\alpha, \beta) \in \mathbb{C}^2$$

ii. Eliminate by the resultant of two quadratics (just use hand computation enough).

iii. (The trick for Σ_2 , from Ilya) If $\deg F = 3$, then

$$\lambda \in \Sigma_2 \iff F(x, y, \lambda) = Q_1 Q_2 + \alpha , \quad \deg Q_1 = 1, \deg Q_2 = 2$$

Part II. our Hamiltonian in Zeeman's form

Our Hamiltonian is (MISSION 2) :

$$H = -\frac{1}{2}(x^2 + y^2) - \frac{B_2}{3}x^3 + B_2xy^2 + (B_1 - A_1)x^2y - \left(\frac{B_1}{3} + A_1\right)y^3$$

We can write it in the complex form by lemma 3(i) :

$$H = -\frac{1}{2}|z|^2 + \operatorname{Re}(\alpha z^3 + \beta z^2 \bar{z}) , \quad \text{with } \alpha = -\frac{B_2 + iB_1}{3} , \quad \beta = iA_1$$

In polar coordinates is :

$$H = -\frac{1}{2}r^2 + r^3h(\theta) , \quad h(\theta) = \operatorname{Re}(\alpha e^{3i\theta} + \beta e^{i\theta})$$

From MISSION 3 we have :

$$\Sigma_1 : \exists \theta , h'(\theta) = h''(\theta) = 0 ; \quad \Sigma_2 : \exists \theta_1 \neq \theta_2 , h'(\theta_1) = h'(\theta_2) = 0 , h(\theta_1) = h(\theta_2) > 0$$

Part III. Hand computation of Σ_1

Step 1 : $h' = 0$ and $h'' = 0$ in complex form

Let $u = e^{i\theta}$. Since $h = \operatorname{Re}(\alpha u^3 + \beta u)$ we have :

$$h' = \operatorname{Re}(i(3\alpha u^3 + \beta u)) , \quad h'' = -\operatorname{Re}(9\alpha u^3 + \beta u)$$

Therefore :

$$h'(\theta) = 0 \iff P := 3\alpha u^3 + \beta u \in \mathbb{R} , \quad h''(\theta) = 0 \iff Q := 9\alpha u^3 + \beta u \in i\mathbb{R}$$

Step 2 : simplify in one complex equation

$$P \in \mathbb{R}, Q \in i\mathbb{R}$$

$\Rightarrow$

$$\begin{cases} P = \bar{P} \\ Q = -\bar{Q} \end{cases}$$

$\Rightarrow$

$$9\alpha u^3 = \bar{\beta}\bar{u} - 2\beta u$$

Multiplying by u and setting $w := u^2$ ($|w| = 1$) :

$$(\star) \quad 9\alpha w^2 + 2\beta w = \bar{\beta} \quad (|w| = 1)$$

Step 3 : eliminate w by the resultant

Taking the conjugate of $(\star)$ and multiplying by w^2 we get two quadratics with a common root :

$$P_1(w) = 9\alpha w^2 + 2\beta w - \bar{\beta}, \quad P_2(w) = -\beta w^2 + 2\bar{\beta}w + 9\bar{\alpha}$$

Resultant of two quadratics polynomials $a_2w^2 + a_1w + a_0$, $b_2w^2 + b_1w + b_0$ is :

$$(a_2b_0 - a_0b_2)^2 - (a_2b_1 - a_1b_2)(a_1b_0 - a_0b_1)$$

Here :

$$\begin{cases} a_2b_0 - a_0b_2 = 81|\alpha|^2 - |\beta|^2 \\ a_2b_1 - a_1b_2 = 18\alpha\bar{\beta} + 2\beta^2 \\ a_1b_0 - a_0b_1 = \overline{18\alpha\bar{\beta} + 2\beta^2} \end{cases}$$

Then :

$$Res(P_1, P_2) = 0$$

$\Rightarrow$

$$(\star\star) \quad (81|\alpha|^2 - |\beta|^2)^2 = |18\alpha\bar{\beta} + 2\beta^2|^2$$

Remark :

Two polynomial f, g have at least one same root $\iff Res(f, g) = 0$

Step 4 : back to (A_1, B_1, B_2)

Substituting $\alpha = -\frac{B_2 + iB_1}{3}$, $\beta = iA_1$:

$$81|\alpha|^2 - |\beta|^2 = 9(B_1^2 + B_2^2) - A_1^2, \quad 18\alpha\bar{\beta} + 2\beta^2 = -2A_1(3B_1 + A_1) + 6iA_1B_2$$

So $(\star\star)$ becomes :

$$(9(B_1^2 + B_2^2) - A_1^2)^2 = 4A_1^2(3B_1 + A_1)^2 + 36A_1^2B_2^2$$

simplify
 $\Rightarrow$

$$\Sigma_1 : A_1^4 + 8A_1^3B_1 + 18A_1^2B_1^2 + 18A_1^2B_2^2 - 27B_1^4 - 54B_1^2B_2^2 - 27B_2^4 = 0$$

Exactly the equation in MISSION 3.

Part IV. Hand computation of Σ_2 (the factorization trick)

In this part , we will use the trick that Ilya mentioned.As follows :

Theorem :

If we suppose our $F(x, y, \lambda)$ satisfies $\deg(F) = 3$

Then we have :

$$\lambda \in \Sigma_2 \iff F(x, y, \lambda) = Q_1 Q_2 + \alpha$$

where

Q_1 is polynomial with $\deg(Q_1) = 1$

Q_2 is polynomial with $\deg(Q_2) = 2$

$\alpha := F(x_1, y_1) = F(x_2, y_2)$ with $(x_1, y_1), (x_2, y_2)$ balanced points

matching coefficients

Set $Q_1 = y - mx - c$ and $Q_2 = q_{20}x^2 + q_{11}xy + q_{02}y^2 + q_{10}x + q_{01}y + q_{00}$. Matching $H - \alpha = Q_1 Q_2$

We check the degree between $H - \alpha$ and $Q_1 Q_2$:

I. the cubic part :

$$(B_1 + 3A_1)m^3 - 3B_2m^2 - 3(B_1 - A_1)m + B_2 = 0 \quad (\text{vi})$$

II. the quadratic , linear and constant parts :

$$B_2(1 - 3m^2) = m(3 - m^2)(B_1 - A_1) \quad (\text{v})$$

CASE $m = 0$: (vi) and (v) gives : $B_2 = 0$

CASE $m \neq 0$: (v) and (vi) gives : $A_1 = 0$

$$i.e. \Sigma_2 \subseteq \{A_1 = 0\} \cup \{B_2 = 0\}$$

Then , we have simplified it in MISSION 3 , that is :

$$\Sigma_2 : A_1 = 0 \text{ or } (B_2 = 0, (A_1 - B_1)(A_1 + 3B_1) < 0)$$

Exactly the equation in MISSION 3.

Part V. Conclusion

The bifurcation diagram of $H(x, y, (A_1, B_1, B_2))$ is :

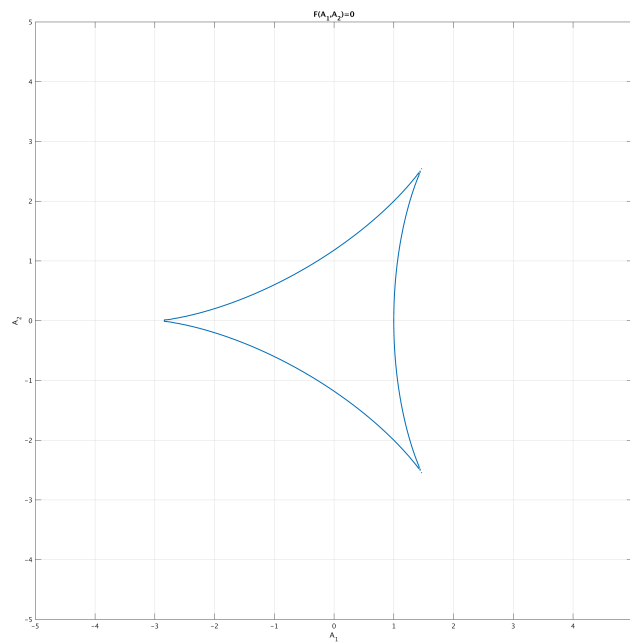
$$\begin{cases} \Sigma_1 : A_1^4 + 8A_1^3B_1 + 18A_1^2B_1^2 + 18A_1^2B_2^2 - 27B_1^4 - 54B_1^2B_2^2 - 27B_2^4 = 0 \\ \Sigma_2 : A_1 = 0 \text{ or } (B_2 = 0, (A_1 - B_1)(A_1 + 3B_1) < 0) \end{cases}$$

This time is totally by hand , NO COMPUTER SOFTWARE !

Moreover , for $H(x, y, (A_1, A_2))$ [The case for $B_1 = 1, B_2 = 0$] is :

$$\begin{cases} \Sigma_1 : & -27 + 18A_1^2 + 8A_1^3 + A_1^4 + 18A_2^2 - 24A_1A_2^2 + 2A_1^2A_2^2 + A_2^4 = 0 \\ \Sigma_2 : & A_2(3A_1^2 - A_2^2) = 0 \end{cases}$$

$\Sigma_1(Caustic)$:



$\Sigma_2(Maxwell)$:

